

Technosphere Collapse and Technosignature Duty Cycles

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1 Abstract

TBD.

2 Introduction

One of the most puzzling questions in astrobiology is whether intelligent civilizations exist elsewhere in the galaxy, and if so, why we have not yet detected them. This apparent contradiction, often referred to as the Fermi Paradox or the Great Silence, raises the question of why, in a galaxy billions of years old, we observe no clear signs of advanced extraterrestrial life. Many solutions have been proposed to explain this Great Silence [Webb, 2010]. One class of explanations suggests that intelligent life is intrinsically rare, perhaps requiring an improbable convergence of planetary, chemical, or biological conditions [Wesson, 1990]. Another contends that while intelligence may emerge, technological civilizations are typically short-lived, succumbing to self-destruction through ecological collapse, conflict, or other systemic instabilities [Vinn, 2024].

The Drake Equation provides a widely used framework for thinking about this question, estimating the number of detectable civilizations (N) based on a chain of astrophysical, biological, and sociotechnical probabilities [Drake, 1965]. Among its terms, the average duration that a civilization emits detectable technosignatures (L) carries disproportionate weight, yet remains the most uncertain. Indeed, even in early discussions of the Drake Equation, it was recognized that, given plausible values for the other terms, N is essentially proportional to L . As a result, even modest changes or uncertainties in L can shift estimates of N by orders of magnitude.

Traditionally, L has been treated as a fixed, continuous span of time; once a civilization switches ON (begins emitting technosignatures), it remains detectable until it either dies out or somehow becomes OFF. This continuous-emission assumption is convenient for Drake Equation calculations, but it is likely unrealistic. Societies may experience turbulence due to resource depletion, climatic upheavals, or internal crises, interrupting their technological activities long before an ultimate end. Historical studies of Earth’s civilizations, for example, show cycles of rise and fall with typical durations on the order of centuries [Shermer, 2002]. It is conceivable that extraterrestrial technological civilizations have similar boom-bust cycles, rather than one long, unbroken broadcast. In fact, it has been cautioned that the Drake Equation tacitly assumes independent and continuously broadcasting civilizations, thereby neglecting important temporal and evolutionary effects [Cirković,

2004]. More recent statistical models confirm that the timing and duration of communicative phases strongly affect detection chances. That is, if civilizations emerge at random times and have finite lifespans, only a small fraction would overlap with our own era, and those overlaps set the probability of contact [Balbi, 2018]. In other words, even if intelligence frequently arises, most civilizations might simply miss each other in time.

It has also been proposed that sustainability and growth limits might curtail L . The “Sustainability Solution” to the Fermi Paradox argues that exponential expansion is unsustainable, so advanced societies may stabilize or collapse before spreading across the Galaxy [Haqq-Misra and Baum, 2009]. In this view, many civilizations could exist without obvious technosignatures if they either limit their growth or suffer periodic setbacks that reset their development. Such civilizations might intermittently go quiet, never achieving the continuous, galaxy-wide presence assumed in some Fermi paradox arguments. These ideas echo the concept of a Great Filter, wherein one or more bottlenecks (perhaps self-inflicted) drastically reduce the number of long-lived, continuously detectable civilizations. The possibility that L is not a single fixed interval but rather composed of active and inactive periods has significant implications. It suggests that searches for technosignatures need to account for duty cycle (i.e., the fraction of time a civilization is actually “on the air”). Indeed, SETI practitioners have begun considering intermittent signals in their strategies. For instance, it has been proposed that an extraterrestrial beacon could conserve energy by transmitting in brief bursts or only at certain times, which would require us to observe a target for longer durations to catch a transmission [Gray, 2020]. This aligns with our motivation to redefine L in terms of an effective duty cycle. Instead of assuming that once a signal appears it will always be present, we ask: what if technosignature emission is episodic? By framing longevity as a duty cycle, we can incorporate the reality of potential off periods (collapse, dormancy, or deliberate quietude) into the Drake Equation. This approach lets us explore a richer set of possibilities, from civilizations that burn brightly but briefly, to those that cyclically renew themselves, to the rare cases that achieve sustainable continuity.

3 Model Description

A framework recently proposed outlines ten plausible futures for Earth-originating civilization 1000 years from now, structured around variations in governance type, resource regime, and systemic resilience [Haqq-Misra et al., 2025]. These scenarios are grounded in Earth’s present-day sociotechnical conditions and extrapolated using techniques from futures studies, including PEST analysis (Political, Economic, Social, Technological), cross-consistency assessment, and structured scenario worldbuilding. Although the framework explores plausible civilizational trajectories, it does not explicitly model the temporal dynamics of collapse and recovery, nor their implications for the persistence and detectability of technosignatures. This leaves open a crucial question: how do different sociotechnical paths shape the *ON/OFF* dynamics of technosignature production (and potential detection)?

To explore this, we build a dynamic model that simulates long-term evolution of technological capacity and resource stocks under varying governance structures, resource regimes, and hazard exposure. Collapse and recovery cycles emerge naturally from feedback between growth and depletion, combined with stochastic existential risk. We use this model to explore how long civilizations remain detectable, how often they go silent, and what kinds of futures might dominate the galactic landscape. From our simulations, we compute the average duty cycle of each scenario, and from this, we calculate an effective detectability duration (L_{eff}), allowing us to assess not just how long

civilizations last, but how often we might catch them in the act of being detectable.

3.1 Simulation framework

We developed a hybrid deterministic-stochastic model to simulate the long-term behavior of technological civilizations, with the goal of capturing collapse-recovery dynamics and their effect on technosignature detectability. This model extends the scenario-based framework introduced by Haqq-Misra et al. [Haqq-Misra et al., 2025], adding temporal dynamics and explicit feedback between resource depletion, technological growth, and systemic collapse. The simulation spans a fixed period of 1000 years, discretized into yearly time steps. At each step, a civilization is characterized by its technological capacity, $T(t)$, and its available resource stock, $R(t)$. Technological capacity increases linearly each year according to a fixed growth rate r , which we take directly from the values reported in Haqq-Misra et al., Table 9. Simultaneously, the available resource stock is depleted at a constant rate δ . If the resource level falls to zero (i.e., $R(t) < 0$), or if an existential risk event occurs (represented as a probabilistic hazard drawn at each time step) the civilization undergoes a collapse. Collapse represents a systemic breakdown in infrastructure, coordination, and technological continuity. It leads to a sharp reduction in technological capacity, determined by a collapse factor c_f , which defines the proportion of technology that survives the event. This loss can be interpreted as the result of disruptions to power grids, communication networks, knowledge retention, or the abandonment of complex systems.

Following collapse, the system enters a dormant recovery period of duration r_d . During this phase, the society is unable to grow or consume resources, reflecting a temporary halt in technological and economic activity. This delay allows time for institutional reorganization, population stabilization, or environmental recovery. Once the recovery period concludes, the resource stock is partially replenished to a fraction r_f of its original value R_0 . This recovery fraction represents retained infrastructure, stored reserves, or ecological regeneration. A full recovery ($r_f \approx 1$) implies robust contingency planning and strong institutional memory, while lower values indicate more severe losses and diminished rebuilding capacity. This cycle of growth, collapse, dormancy, and recovery can repeat multiple times during the simulation window, depending on the fragility of the system, the frequency of stochastic hazards, and the depletion rate of resources.

Formally, let the initial conditions be $T_0 = 1$, and $R_0 \in \mathbb{R}_+$, and define the indicator function as $\chi(t) \in \{0, 1\}$, with $\chi(t) = 1$ if the system is active (not collapsed or recovering) at time t , and $\chi(t) = 0$ otherwise. The update rules for each time step are given by

$$\begin{aligned} T_{t+1} &= \chi_t (T_t + r) + (1 - \chi_t) T_t, \\ R_{t+1} &= \chi_t (R_t - \delta) + (1 - \chi_t) R_t. \end{aligned} \tag{1}$$

Equations (1)–(3) define the full dynamical cycle of growth, collapse, and recovery in the model.

We include both a deterministic and a stochastic trigger for collapse to reflect two distinct failure modes: internal depletion and external shocks. Collapse is deterministically triggered when the resource stock is fully depleted (i.e., $R(t) \leq 0$). It may also be triggered by a stochastic existential hazard, which we model as a Bernoulli process. At each time step t , a uniform deviate $U(t) \sim U(0, 1)$ is drawn. If $U(t) < h$, where $h \in [0, 1]$ is the hazard rate specific to the scenario, then a collapse is triggered independently of resource levels.

Upon collapse, the system state is reset according to

$$\begin{aligned} T_{t+1} &= c_f T_t, \\ R_{t+1} &= 0. \end{aligned} \tag{2}$$

The system then remains inactive for r_d time steps, after which resource availability is restored as

$$R_{t+r_d} = r_f R_0. \tag{3}$$

We define a civilization as being ON when it maintains a positive level of resources (i.e., $R(t) > 0$). This assumption reflects the idea that some minimal material throughput is necessary for societal function and detectability, including energy consumption, communication, and technosignature production. Conversely, when resources are fully depleted ($R(t) = 0$), the civilization is considered OFF, representing a dormant, collapsed, or otherwise undetectable state. This binary definition underlies the calculation of duty cycle and related metrics discussed in subsection 3.3.

3.2 Scenario typologies and parameter mapping

We apply our model to the ten future scenarios to simulate the long-term behavior of civilizations under diverse sociotechnical conditions. Each scenario describes a distinct vision for humanity’s trajectory, defined by combinations of resource abundance, governance structure, institutional resilience, and technological integration. Our goal is to translate these qualitative narratives into a coherent set of quantitative parameters that govern dynamics of growth, collapse, and recovery. Scenarios are characterized along three conceptual axes: resource regime (scarcity vs. non-scarcity), governance type (from autocratic to participatory), and structural design (hierarchical vs. distributed). Scenarios are also mapped onto six thematic clusters (Table 1).

We assign parameter values based on the sociotechnical features of each scenario. The growth rate r is taken directly from Haqq-Misra et al. (2025) and held fixed. **We note that assuming technological capacity increases linearly each year via r is a simplification, especially given scenarios that imply discontinuities such as collapse points or radical transformation points; we retain it here for consistency with the source parameterization.** The remaining parameters (initial resource stock R_0 , per-year depletion rate δ , collapse depth c_f , recovery delay r_d , recovery fraction r_f , and existential hazard rate h) are derived from the narrative structure of each scenario and converted into quantitative values through structured reasoning.

We assume that governance structure influences how technological civilizations respond to systemic stress and collapse. Highly centralized systems (rule by one) could be more vulnerable to severe and prolonged disruptions, as decision-making and critical infrastructure are concentrated and potentially less adaptable. When collapse occurs (due to resource exhaustion or external hazard) such civilizations may face steeper declines in technological capacity and slower recovery, particularly if institutional continuity is fragile. In contrast, more distributed governance (rule by all) might support greater redundancy and flexibility, enabling shallower collapses and faster recovery following disruption. To reflect these hypothesized patterns, we selected representative parameter value ranges for collapse depth c_f , recovery delay r_d , and recovery fraction r_f that align with the governance structure assumed in each scenario (Table 2). These values are not based on empirical observation, but represent internally consistent assumptions designed to explore contrasting trajectories in our scenario-based simulations.

Table 1: Scenario descriptions and sociotechnical tags

Scenario	Tags	Description
S1: Big Brother is Watching	Scarcity + Rule by One + Hierarchical Cluster 6 (Earth 2024)	Centralized authoritarian system under resource stress
S2: Wild West	Scarcity + Rule by Few + Hierarchical Cluster 6 (Earth 2024)	Deregulated, competitive, and fragmented society
S3: Golden Age	Non-scarcity + Rule by All + Distributed Cluster 6 (Earth 2024) [†]	Stable egalitarian society with abundant resources
S4: Living with the Land	Scarcity + Rule by Few + Hierarchical Cluster 5 (Simplicity)	Low-impact, cyclical, ecologically constrained society
S5: Transhumanism	Non-scarcity + Rule by All + Distributed Cluster 4 (Engineering)	Technologically integrated, post-biological civilization
S6: Sword of Damocles	Scarcity + Rule by Few + Hierarchical Cluster 4 (Engineering)	High-tech and high-risk system with cascading failure potential
S7: Restoration	Non-scarcity [‡] + Rule by All + Distributed Cluster 3 (Sustainability)	Recovered post-collapse world with resilience
S8: Ouroboros	Scarcity + Rule by Few + Hierarchical Cluster 3 (Sustainability)	Oscillatory civilization with cyclical collapse and regrowth
S9: Deus Ex Machina	Non-scarcity + Rule by All + Distributed Cluster 2 (Tech Escape)	Aggressive technological expansion with instability risk
S10: Out of Eden	Non-scarcity + Rule by All + Distributed Cluster 1 (Parallel Life)	Harmonized coexistence with nature and machines

[†]Cluster assignments follow [Haqq-Misra et al. \[2025\]](#); S3 shares the Earth 2024 technology cluster with S1 and S2 despite differing in global factors.

[‡]S7 is tagged non-scarcity in the original framework but parameterized with scarcity-level resource pressure to model post-collapse recovery (see text).

One exception is the Restoration scenario (S7). While it is categorized as non-scarcity and governed by distributed institutions, it is conceptually framed as a rebuilding civilization emerging from an earlier collapse. To reflect this, we initialize the simulation with a reduced effective resource stock and a moderate depletion rate (parameters that would otherwise suggest scarcity), thereby modeling a society recovering from prior systemic failure. This allows us to represent S7 as a post-collapse recovery regime without forcing a collapse to occur within the simulation window.

The Ouroboros scenario (S8) is treated differently. In the original framework, S8 is described as a civilization that repeatedly undergoes expansion, collapse, and regrowth, with the year 3000 situated partway through an ongoing recovery phase. Rather than imposing externally timed or periodic collapse events, we model this behavior as an emergent dynamical regime. Specifically, S8 is assigned moderate technological growth, partial technological survival during collapse, and a relatively high recovery fraction of planetary resources. This combination allows repeated collapse–recovery cycles to arise endogenously from the interaction between resource depletion and rebuilding, and permits exploration of whether oscillatory behavior remains stable or dampens over long timescales.

Similarly, Deus Ex Machina (S9) is tagged as non-scarcity in the original framework but is parameterized with moderate resource pressure ($R_0 = 900$, $\delta = 1.3$) to reflect the high extraction demands of its aggressive technological expansion program, which drives resource depletion despite the scenario’s nominally post-scarcity classification.

Table 2: Mapping sociotechnical structure to parameter logic

Governance Type	Collapse Depth c_f	Recovery Delay r_d	Recovery Fraction r_f
Rule by One	0.5–0.7	40–60	0.10–0.20
Rule by Few	0.2–0.6	50–100	0.10–0.35
Rule by All*	0.3–1.0	0–70	0.20–1.0

*Post-scarcity idealizations (S3, S10) use $c_f = 1.0$, $r_d = 0$, $r_f = 1.0$, reflecting no-collapse conditions. S7 and S9 use parameters outside the typical governance range as narrative exceptions (see text).

Scenarios also differ fundamentally in their assumptions about planetary resource conditions. Rather than define scarcity solely in terms of starting resources, we interpret it as a function of resource pressure (i.e., how quickly a civilization depletes its reserves). That is, systems with a large initial stock may still experience scarcity if it depletes resources at a high rate (e.g., S6). Conversely, a civilization with a modest stock may persist through repeated collapse–recovery cycles if post-collapse resource recovery partially replenishes its base (e.g., S8, S9), or may rebuild following an earlier systemic failure (e.g., S7).

Table 3: Interpreting resource pressure from R_0 and δ

Resource Pressure	R_0	δ	Scenarios
High	≤ 1000	1.0–2.5	S1, S2, S4, S7 , S8, S9
Moderate	400–1600	0.5–1.0	S6
Low	≥ 1200	0.0–0.5	S3, S5, S10

We implement Monte Carlo simulations to explore how uncertainty in model parameters shapes the long-term behavior of each scenario. Each scenario’s baseline values (Table 4) are perturbed using probabilistic variation to capture plausible uncertainty (Table 5). Initial resource stock (R_0)

and depletion rate (δ) are sampled from normal distributions with modest standard deviations to represent uncertainty in ecological baselines and consumption patterns. Collapse depth (c_f) and recovery fraction (r_f) are sampled using triangular distributions centered on scenario-specific values, allowing outcomes to concentrate near expected levels while still capturing asymmetries in resilience and recovery capacity. Recovery delay (r_d) is sampled from a normal distribution with a standard deviation of ± 10 years, reflecting variability in rebuilding pace across different societal structures.

The existential hazard rate (h) is held fixed for each scenario because it encodes an assumed background level of external risk (such as AI misalignment, interstellar conflict, or catastrophic environmental feedbacks) rather than internal system dynamics. Unlike other parameters whose variability arises from structural or operational uncertainty, h reflects a narrative-level judgment about a civilization’s exposure to exogenous threats. Scenarios characterized by unstable geopolitical contexts, risky technological dependencies, or unresolved existential challenges (e.g., S1, S6, S7) are assigned higher hazard rates to reflect this vulnerability, while S5 carries a small residual risk due to high systemic complexity. In contrast, scenarios with societal stability, low-impact lifestyles, or resilient infrastructure (e.g., S3, S4, S8, S9, S10) are assigned zero or near-zero hazard rates. Keeping h fixed rather than sampling it probabilistically allows for clearer attribution of collapse outcomes to either endogenous fragility or exogenous shocks, rather than conflating the two.

Table 4: Final parameter values for each scenario

Scenario	r	R_0	δ	c_f	r_d	r_f	h
S1	0.0020	400	1.2	0.6	40	0.15	0.003
S2	0.0026	700	1.5	0.3	70	0.10	0.001
S3	0.0001	2400	0.0	1.0	0	1.0	0.0
S4	0.0060	600	2.5	0.2	100	0.10	0.0
S5	0.0003	2500	0.1	0.9	20	0.80	0.0005
S6	0.0018	1600	1.0	0.4	50	0.25	0.005
S7	0.0060	400	1.5	0.3	60	0.20	0.001
S8	0.0015	800	1.3	0.6	70	0.35	0.0
S9	0.0011	900	1.3	0.5	70	0.35	0.0
S10	0.0002	2700	0.0	1.0	0	1.0	0.0

Table 5: Parameter uncertainty assumptions for Monte Carlo sampling

Parameter	Distribution	Uncertainty
R_0	Normal	$\pm 5\%$ of mean
δ	Normal	$\pm 5\%$ of mean
c_f	Triangular	$\pm 5\%$ of mode
r_d	Normal	± 10 years
r_f	Triangular	$\pm 5\%$ of mode

3.3 Metrics

We define four metrics to reflect different aspects of technospheric persistence and collapse dynamics across Monte Carlo simulations. These metrics are designed to capture the onset, frequency, and

continuity of civilization-level activity, with a focus on their implications for detectability.

The first metric, T_c , is the mean time to first collapse across replicates. It serves as a proxy for system fragility: scenarios with low values of T_c tend to break down early, while those with higher values demonstrate greater initial stability. When a simulation run does not experience collapse within the 1000-year window, the full timespan is used. The second metric, f_{nc} , denotes the fraction of runs that avoid collapse entirely. This value reflects the overall robustness of a scenario and helps differentiate consistently stable futures from those vulnerable to systemic failure. We also measure D_c , the average duty cycle, defined as the fraction of time a civilization remains active, that is, maintaining nonzero resource levels and thus potentially emitting technosignatures. This definition assumes that minimal resource availability is a prerequisite for detectable activity. Scenarios with high D_c maintain persistent activity, while those with low D_c may cycle through long dormant periods or experience early termination. Finally, N_c , the mean number of collapse-recovery cycles, captures whether a scenario tends to exhibit singular collapse events or undergo repeated disruptions over time. Together, these metrics provide a compact but informative summary of each scenario’s trajectory, enabling comparisons across sociotechnical regimes and offering insight into the reliability and intermittency of technosignature emission.

4 Results

4.1 Monte Carlo Outcomes and Temporal Profiles

We ran 200 independent simulations for each scenario, introducing probabilistic variation in model parameters as described in Section 3. Figure 1 shows the resulting trajectories of global technology $T(t)$ and available resources $R(t)$, averaged across runs. These profiles reveal the distinct collapse and recovery dynamics that characterize each scenario. As expected, Golden Age (S3) and Out of Eden (S10) show uninterrupted growth and resource stability throughout the 1000-year window, consistent with their post-scarcity narratives and reflected in perfect duty cycles ($\bar{D}_c = 1.0$) and no collapse events ($\bar{N}_c = 0$). In stark contrast, scenarios such as Big Brother (S1) and Sword of Damocles (S6) suffer early collapses ($\bar{T}_c < 210$) and high collapse frequency ($\bar{N}_c \approx 4.5$ for S6 and nearly 10 for S1), leading to sharp contractions in technological capacity and prolonged periods of inactivity. Deus Ex Machina (S9), although collapsing later ($\bar{T}_c \approx 689$), also experiences multiple interruptions and never recovers full continuity, with only modest duty cycle ($\bar{D}_c \approx 0.91$) and about 1.5 collapse events on average.

More complex dynamics emerge in intermediate cases. Transhumanism (S5) avoids collapse in roughly two-thirds of runs and achieves a near-continuous duty cycle ($\bar{D}_c \approx 0.99$), albeit with slower growth and occasional extinction. Living with the Land (S4), in contrast, undergoes frequent and deep collapses ($\bar{N}_c \approx 6.6$), producing the lowest sustained activity among all non-terminal scenarios ($\bar{D}_c \approx 0.38$) despite a steep initial growth phase. Restoration (S7), designed as a post-collapse recovery regime, displays consistent rebuilding behavior: early collapses ($\bar{T}_c \approx 223$), followed by approximately 7.5 recovery cycles and moderate continuity ($\bar{D}_c \approx 0.57$). Ouroboros (S8) exhibits a distinct pattern of low-frequency, self-stabilizing oscillations: it collapses only twice on average, yet maintains robust technological levels and a high duty cycle ($\bar{D}_c \approx 0.87$), reflecting a regime that cycles gracefully without exhausting resources or triggering runaway collapse.

Because the S8 narrative implies repeated collapse–recovery cycles, we ask whether those oscillations are always transient or can be sustained; we probe this with a sweep over r_f and δ (Figure 2).

Panel A shows deterministic technology trajectories for varying r_f while holding all other S8 parameters at their baseline values. For low recovery fractions ($r_f \leq 0.15$), the system undergoes rapid, repeated collapse–recovery cycles (3 cycles within the 1000-year window), as the diminished resource base after each recovery is quickly exhausted. At intermediate values ($0.20 \leq r_f \leq 0.50$), the system sustains two collapse–recovery cycles with longer growth phases between collapses. At higher recovery fractions ($r_f = 0.60$), the first collapse is delayed significantly, and only a single cycle occurs within the simulation window. Panel B maps the number of collapse cycles across the (r_f, δ) parameter space, revealing a clear regime boundary: high depletion rates combined with low recovery fractions produce the most frequent oscillations (up to 8 cycles), while low depletion and high recovery fractions suppress collapse entirely. The baseline S8 parameters (marked with a star) sit near the boundary between 2- and 3-cycle regimes, suggesting that modest shifts in resource recovery capacity could qualitatively alter the civilization’s long-term trajectory. These results confirm that S8’s oscillatory behavior is not universally stable and it depends sensitively on the balance between resource depletion and post-collapse recovery, with some parameter combinations producing sustained cycling and others leading to dampened or terminal collapse.

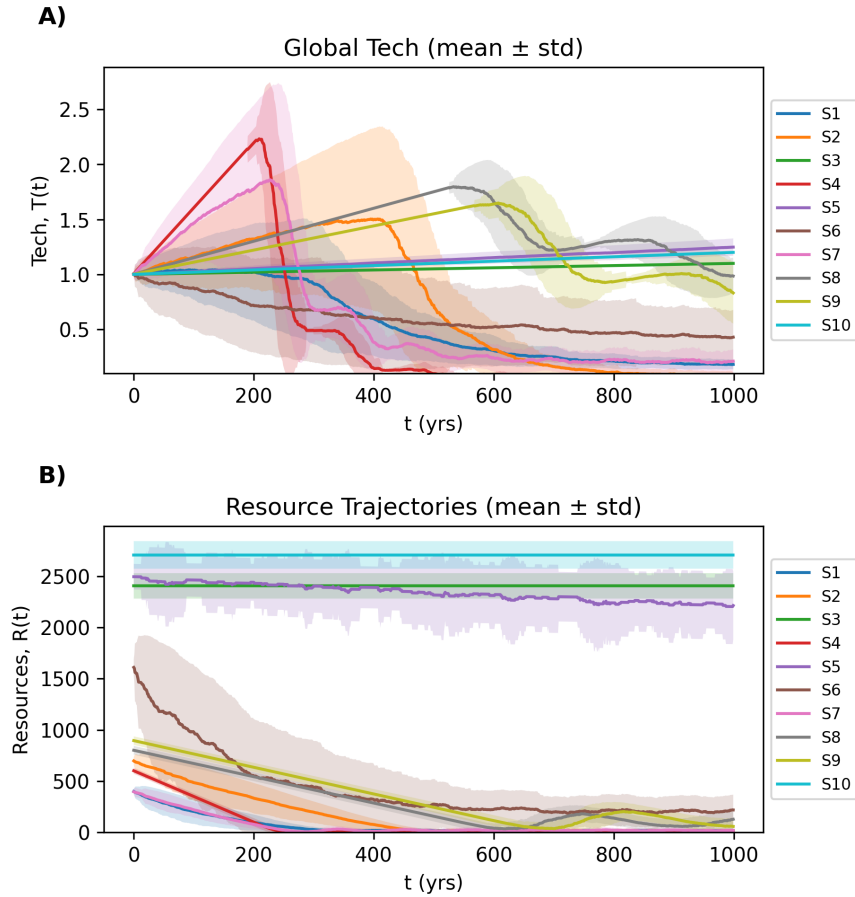


Figure 1: Temporal dynamics of simulated civilizations across ten scenarios. (A) Mean trajectories of global technology $T(t)$ with ± 1 standard deviation shaded. (B) Mean resource levels $R(t)$ with ± 1 standard deviation shaded.

Figure 3 summarizes the outcomes of these simulations using the average of the four metrics described above ($\bar{T}_c$, f_{nc} , $\bar{D}_c$, $\bar{N}_c$). While Golden Age (S3) and Out of Eden (S10) remain fully

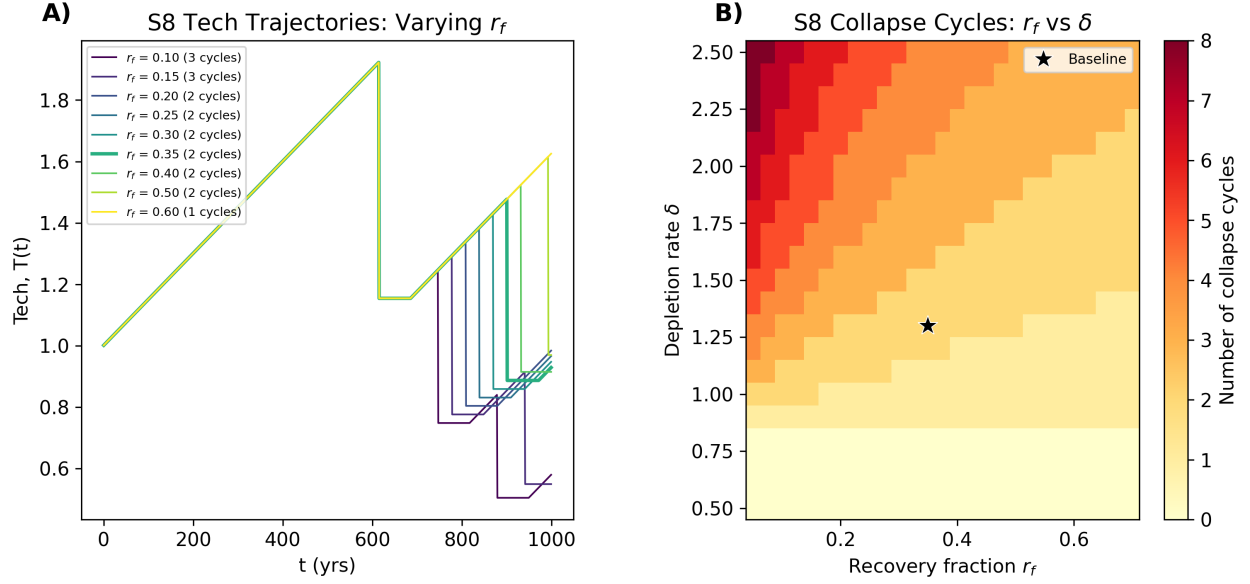


Figure 2: Parameter sensitivity of S8 (Ouroboros) oscillatory dynamics. (A) Deterministic technology trajectories $T(t)$ for varying recovery fraction r_f , with all other parameters held at baseline values. The number of collapse–recovery cycles within the 1000-year window is indicated in the legend. (B) Number of collapse–recovery cycles as a function of recovery fraction r_f and depletion rate δ . The star marks the baseline S8 parameter combination.

stable throughout the 1000-year window, with no collapses and continuous activity ($f_{nc} = 1.0$, $\overline{D}_c = 1.0$), other futures display more volatile dynamics. **Big Brother (S1)** and **Sword of Damocles (S6)** show early collapses ($\overline{T}_c \leq 220$) and high collapse frequency, while **Deus Ex Machina (S9)** collapses later ($\overline{T}_c \approx 689$) but still undergoes multiple recovery cycles, reflecting fragile systems with intermittent technosignature potential.

4.2 Implications for the Drake Equation and the Great Silence

The results of our simulations have direct implications for the Drake Equation, which offers a widely used framework for estimating the number of detectable civilizations in our galaxy as

$$N = R_* f_p n_e f_l f_i f_c L. \quad (4)$$

Each parameter represents a step in the emergence of detectable life: R_* is the rate of star formation, f_p the fraction of those stars with planets, n_e the number of habitable planets per system, f_l the fraction of planets where life arises, f_i the fraction where intelligence evolves, f_c the fraction that develop detectable technology, and L the average duration such civilizations remain detectable. While early terms are rooted in astrophysical and biological probabilities, the final term L carries outsized importance in addressing the Fermi Paradox. If civilizations exist but are short-lived or frequently silent, N could remain small despite favorable conditions for life elsewhere.

Traditionally, L is treated as a static, continuous span: once a civilization becomes detectable, it remains so until it disappears. However, our simulations suggest that this assumption dramatically oversimplifies the lived dynamics of sociotechnical systems. Many of our modeled futures exhibit

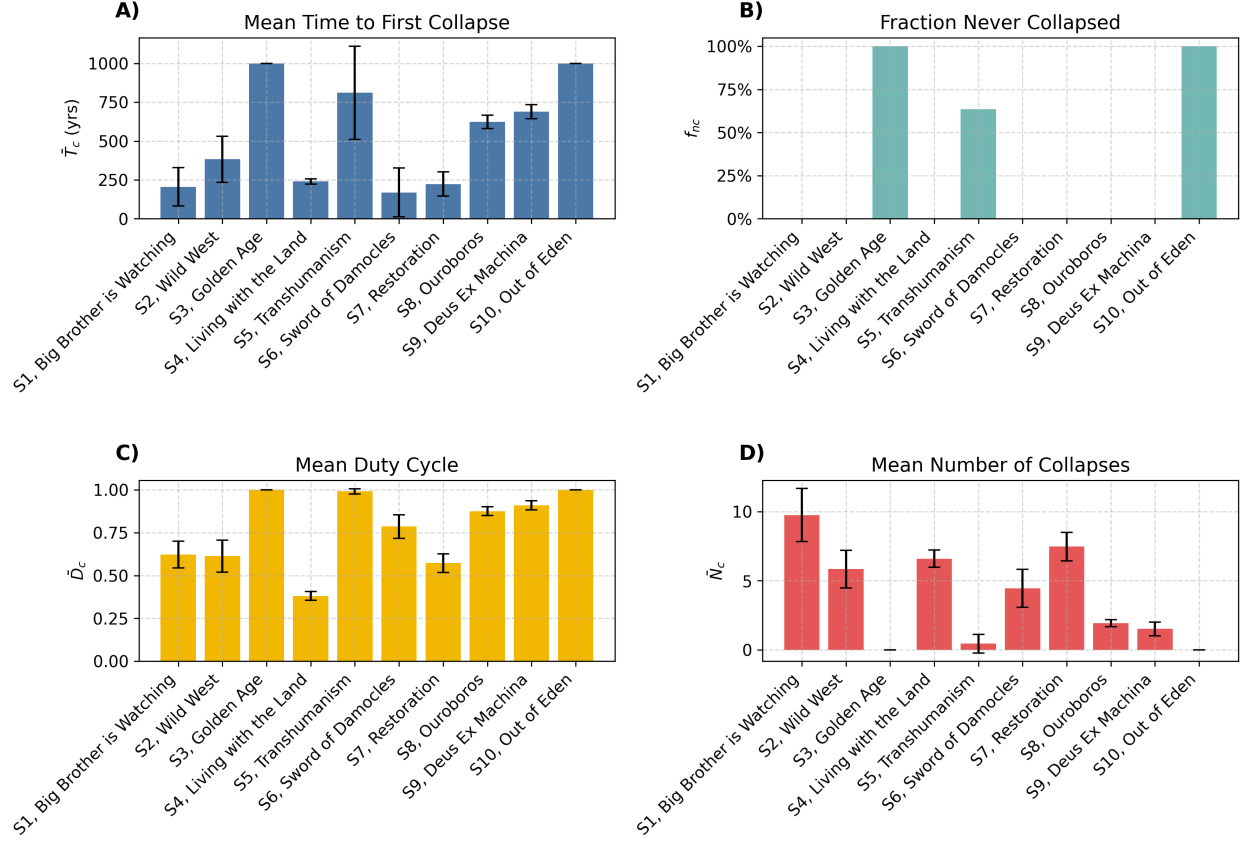


Figure 3: Distributions of key collapse and recovery metrics across scenarios (200 runs each). (A) Mean time to first collapse $\bar{T}_c$ with standard deviation bars. (B) Fraction of simulations that never experienced collapse f_{nc} . (C) Mean duty cycle $\bar{D}_c$, defined as the fraction of time the technosphere remains active. (D) Mean number of collapse events $\bar{N}_c$.

intermittent collapse and recovery cycles, resource-driven dormancy, or stochastic failure due to external hazards. These patterns imply that civilizations may oscillate between ON and OFF phases, producing irregular technosignature signals that are easily missed.

To account for this, we introduce an effective detectability duration,

$$L_{\text{eff}} = D_c T_{\text{span}}, \quad (5)$$

where D_c is the mean duty cycle (defined above as the fraction of time a civilization remains active) and T_{span} is the observation window (1000 years in our simulations). Some have also proposed a related refinement to Drake’s equation in the form of a “reappearance factor” for planets, where civilizations can arise multiple times in succession [Bhattacharya et al., 2011]. For example, even if one technological society collapses after 10,000 years, life may persist on that planet and eventually give rise to a new intelligent species. This factor differs from duty cycle in that it counts sequential independent civilizations over geological timescales, whereas duty cycle refers to the active vs. inactive phases within a single civilization’s lifespan.

Among the ten scenarios, D_c spans a wide range in our simulations. In the most collapse-prone scenario, S4, D_c was as low as 0.381, indicating these civilizations spend well over half of their time in a collapsed or dormant state. By contrast, scenarios S3 and S10 had $D_c = 1.00$,

never experiencing a collapse (continuous activity). Most scenarios fall somewhere in between. For example, S5 has a high $D_c \approx 0.99$, whereas S2 reaches only $D_c \approx 0.61$, reflecting prolonged instability. This variability in D_c has direct implications for the detectability of civilizations. Even if intelligent life is widespread, the duty cycle reduces the probability of intercepting a signal at a given moment. Civilizations may exist for millennia, but if they emit only intermittently, the chances of overlap with our own observational window shrink significantly.

The simulation results also raise a deeper question about the distribution of civilizations in the galaxy: should we expect them to be uniform in character and behavior, or drawn from a more diverse ensemble of trajectories? A uniform distribution, where all civilizations resemble a stable high-duty scenario like S3, would imply a galaxy populated by long-lived, continuously emitting technospheres. Such a distribution would make the silence we observe profoundly puzzling. On the other hand, if the distribution is non-uniform, with most civilizations following fragile, collapse-prone paths (e.g., S1, S2, S6), then long-duty-cycle civilizations would be statistical outliers. In this case, silence becomes more plausible, not because civilizations are rare, but because most are quiet or transient for the vast majority of their lifespan.

Our findings support the latter view. Out of 200 simulations per scenario, only S3 and S10 avoided collapse entirely ($f_{nc} = 1.00$). Most other scenarios saw high collapse frequency (e.g., $\bar{N}_c = 9.76$ for S1), and moderate or low duty cycles. These collapse-prone scenarios suggest that many civilizations may not follow monotonic energy-growth trajectories. Instead of ascending smoothly up the Kardashev scale, they may oscillate or even regress, casting doubt on the assumption that energy use (and thus detectability) increases predictably over time. This indicates that robustness might not be the norm. Civilizations that achieve continuous detectability, such as those in post-scarcity or ecologically harmonized futures, appear rare and structurally exceptional. If galactic civilizations follow a similar distribution, then the Great Silence may simply reflect the dominance of fragile or intermittently silent technospheres.

In this light, the silence we observe may not contradict the possibility of intelligent life, rather, it may reflect the statistical weight of civilizations that emit briefly or not at all during our observational epoch. A non-uniform distribution with many S1/S2-type worlds and very few S3/S10-type worlds reconciles the Drake Equation with current data, while supporting the need for SETI strategies that target not just strong, continuous signals but also faint, episodic, or relic technosignatures.

4.3 Persistence of Technosignatures Beyond Civilizational Activity

The effective detectability term we propose, L_{eff} , captures the active fraction of a civilization’s technological lifespan (i.e., the periods when it is ON and producing observable outputs). However, this formulation assumes that detectability ends when activity ceases. In practice, many technosignatures may outlast their creators, continuing to be observable well after the civilization itself has collapsed or gone quiet. To account for this, we introduce a persistence term, τ , representing the average duration that technosignatures remain detectable after each *ON* period. The revised detectability window, $L_{\text{eff}}^{(d)}$, becomes

$$L_{\text{eff}}^{(d)} = D_c T_{\text{span}} + N_{\text{ON}} \tau. \quad (6)$$

where N_{ON} is the number of active intervals in the simulation. This extension is particularly relevant for short-lived or intermittent civilizations as long-lasting residual signals could greatly increase detectability even if active periods are brief.

Orbital debris and artificial satellites can last from decades to as long as 10,000 to 100,000 years. Satellites in low Earth orbit typically deorbit within decades due to atmospheric drag, while those in geosynchronous orbit, facing minimal resistance, may persist for much longer. Without maintenance, a dense band of satellites in the so-called Clarke belt would slowly degrade through collisions and orbital perturbations over tens of thousands of years, eventually becoming undetectable [Socas-Navarro, 2018]. Megastructures such as Dyson spheres, space mirrors, or stellar engines may persist longer still, though their survival depends on structural design and orbital stability. Rigid constructs without active station-keeping may collapse or drift within a few centuries to several thousand years [Wright, 2020], however, more modular configurations, like swarms of solar collectors, could remain intact for up to hundreds of thousands or even millions of years if not disrupted by collisions.

Atmospheric technosignatures provide the shortest windows of detectability. On Earth, CFC-11 and CFC-12 have atmospheric lifetimes of about 55 and 140 years, respectively, before being broken down by photolysis or other chemical processes [Unk]. Other compounds are more resilient: tetrafluoromethane (CF_4) persists for $\sim 50,000$ years, hexafluoroethane (C_2F_6) for $\sim 10,000$ years, and octafluoropropane (C_3F_8) for $\sim 2,600$ years [Trudinger et al., 2016]. Longer-lived isotopic signatures are conceivable, but their remote detection remains technically challenging. Electromagnetic technosignatures such as radio or laser transmissions propagate indefinitely but are only practically detectable for finite periods. Focused, high-power signals might remain observable for a few hundred light-years over spans of centuries to millennia, depending on the strength of the signal and the sensitivity of receivers [Sheikh et al., 2025].

These persistence times imply that τ can range from negligible (radio bursts) to dominant (orbital infrastructure). In the case of short-lived civilizations with brief ON phases, the contribution of $N_{\text{ON}} \cdot \tau$ may exceed that of $D_c \cdot T_{\text{span}}$. In contrast, for continuously active or long-lived civilizations, L_{eff} dominates and τ contributes relatively little. This distinction matters. If most detectable technosignatures are short-lived (e.g., radio leakage), then D_c and T_{span} remain the key parameters, and our results suggest low duty cycles make detection unlikely. But if many technosignatures persist for millennia beyond collapse, then the silence we observe becomes harder to attribute solely to civilizational fragility. Instead, it would require that either technosignatures are intrinsically ephemeral, or civilizations rarely produce persistent artifacts.

5 Discussion

Our results reframe the Drake Equation’s longevity term in a way that addresses the Fermi Paradox with greater nuance. Rather than requiring that intelligent life be exceedingly rare, the Great Silence may simply indicate that technosignatures are intermittent and timing-dependent [Balbi, 2018]. Aligned with this, recent work suggests that civilizations might commonly face phase transitions or limits to growth that prevent long, unbroken periods of signal emission [Wong and Bartlett, 2022]. In fact, our bimodal outcomes, with some scenarios yielding repeated collapse and others achieving long-term stability, closely echo this hypothesis. The overall idea is that planetary civilizations may follow one of two paths: either “asymptotic burnout”, collapsing under the weight of exponential growth, or a “homeostatic awakening” in which they deliberately curb expansion to achieve equilibrium. In the latter case, a civilization might survive indefinitely but become hard to detect due to a lack of aggressive expansion or high-power signaling. Our high duty-cycle scenarios like Golden Age (S3) or Out of Eden (S10) resemble this stable, sustainable type, whereas the collapse-prone scenarios like Big Brother (S1) or Sword of Damocles (S6) are akin to burnout

cases. The implication is that the rarity of continuous technosignature emitters (high L or D_c) in our simulations could reflect a Galactic trend, where long-lived, broadcasting civilizations might be the exception, not the rule.

An important extension of this idea is that technology can outlast its creators. Our study has focused on the active phase of civilizations (the ON periods), but one must consider that even a civilization which collapses (goes OFF) might leave behind artifacts or long-lived technosignatures. For example, it has been argued that the maximum longevity of a technosignature could be far greater than the civilization’s biological lifespan, because machines, probes, or infrastructure can continue operating (or at least remain observable) for millennia or longer [Wright et al., 2022]. In this view, Earth and humanity may be poor guides to the true value of L , since advanced extraterrestrial technology might persist well beyond the civilization’s demise. This is a crucial point for SETI as even extinct or dormant civilizations could be detectable if they’ve built something durable [Ćirković et al., 2019]. Dyson’s classic idea of searching for waste heat from starlight-absorbing megastructures is a case in point. A “Dyson Sphere” could remain in orbit around a star for eons, glaring in the infrared, even if the society that built it is long gone [Dyson, 1960]. Such relic technosignatures (sometimes called “Dysonian” technosignatures) broaden our notion of detectability. In the same vein, a dense belt of artificial satellites around a planet (a “Clarke exobelt”) might be detectable via its subtle transit signature [Socas-Navarro, 2018]. Other examples include planetary atmospheric anomalies (e.g., chlorofluorocarbon pollution or unusual isotopic ratios in a planet’s spectrum) that could signal past industrial activity [Wright et al., 2022]. In short, not all technosignatures are “live broadcasts”. Some might be fossils or afterglows of past epochs of activity.

Our simulations suggest that the traditional interpretation of the Drake Equation’s longevity term, L , as a continuous span of technosignature emission, may be unrealistic. Across a wide range of plausible futures, we find that civilizational activity is frequently interrupted by collapse, resource depletion, or other disruptions. In most scenarios, the emission of technosignatures is not constant but episodic, raising the need for a revised formulation. We propose an effective detectability duration, $L_{\text{eff}} = D_c \times T_{\text{span}}$, where D_c is the mean duty cycle and T_{span} is the time horizon. This redefinition captures the operational reality of civilizations that switch between active and dormant states. This intermittency reframes how N , the number of detectable civilizations, should be interpreted. If L is better understood as L_{eff} , then models that assume continuous detectability may significantly overestimate N . Conversely, incorporating duty cycle effects helps align theoretical predictions with empirical null results. A galaxy populated by civilizations with low duty cycles would remain largely silent, even if intelligent life is common.

This framework has two important implications. First, it helps explain the Great Silence. Civilizations may exist in abundance, but if they are detectable only intermittently, the likelihood that their technosignatures overlap with our current observational window is low. The apparent absence of signals may not reflect the rarity of extraterrestrial technology, but rather the rarity of sustained emission. Second, it suggests that long-lived, continuously emitting civilizations are outliers. Most of our simulated futures spent significant portions of time in silent or collapsed states, even if they eventually recovered. If such patterns are typical across the galaxy, then detection becomes a matter of timing: for two intermittent civilizations, the probability of mutual visibility is the product of their duty cycles, which is potentially very small. This stands in contrast to the assumptions embedded in the Kardashev scale [Kardashev, 1964], which classifies civilizations by their continuous energy use and implies long-term, uninterrupted detectability. Our simulations instead point to dynamic energy throughput, with civilizations temporarily reaching higher levels,

such as approaching Type I, only to regress during periods of collapse or dormancy. These fluctuations reduce their cumulative detectability and challenge the static expectations of the Kardashev framework. Moreover, while our models focus on collapse-driven intermittency, it's worth noting that some civilizations might limit their energy use deliberately (in pursuit of ecological balance or long-term stability, for example), remaining below Type I not due to collapse, but by design.

Our results also help reconcile seemingly contradictory perspectives on civilizational longevity. One view posits that technological civilizations are fundamentally fragile and prone to self-destruction soon after emergence, driven by ecological, behavioral, or structural instabilities [Vinn, 2024]. Another suggests that once civilizations surpass key thresholds of resilience and coordination, they may persist indefinitely, steadily accumulating across cosmic time [Grinspoon, 2009]. Both patterns emerge naturally in our simulations. Scarcity-driven, hierarchically governed scenarios frequently enter collapse-recovery cycles, resulting in low duty cycles and limited detectability. In contrast, post-scarcity, distributed scenarios maintain stable resource use and continuous activity, supporting high-duty-cycle technospheres. These differences arise not from exogenous shocks, but from internal sociotechnical structure.

Finally, several limitations warrant discussion. Our modeling framework is grounded in Earth-originating scenario typologies, drawing from narrative futures methodologies informed by contemporary geopolitical, ecological, and technological trends. These projections embed assumptions about governance structures, resource constraints, and recovery dynamics that may not extend to non-terrestrial civilizations or to those that do not follow Earth-like developmental pathways. While this grounding enables empirical plausibility, it limits extrapolation beyond our biosphere's trajectory [Rubin, 2001, Denning, 2011]. Moreover, our simulations assume bounded planetary systems and do not account for galactic colonization, interstellar migration, or modes of expansion that could alter the spatial and temporal visibility of civilizations [Walters et al., 1980, Papagiannis, 1983]. We also exclude the effects of cultural convergence, information saturation, or technological transcendence (e.g., the emergence of post-biological intelligence), all of which could suppress or qualitatively change technosignature output. Our operational definition of detectability (based on internal sociotechnical continuity) does not differentiate among specific technosignature classes (e.g., radio emissions, waste heat, megastructures), nor does it address detection thresholds or instrumental sensitivity.

In summary, our results can partially explain the Great Silence: if detectability is often sporadic, then long periods without contact may be the expected baseline. Intermittency may be typical and recognizing that could refine our expectations and expand the scope of SETI.

6 Conclusions

This repository scaffolds a fully reproducible manuscript: figures are programmatically generated, sources are modularized, and builds are automated with `latexmk`. Researchers can now focus on scientific insight rather than plumbing.

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